MATH 350 Advanced Calculus

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August 27, 2025

Outline

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1. Real Analysis Lecture 2
   - Types of real numbers
   - Suprema and Infima

Inductive sets

A subset S of $\mathbb{R}$ is called a **inductive set** if

- $1 \in S$
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Examples:

$$\mathbb{R}, \quad \mathbb{Q}, \quad (0, \infty), \quad \mathbb{Z}, \quad \mathbb{N}, \ldots$$

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$$\mathbb{Q} = \{a/b : a, b \in \mathbb{Z},\ b \neq 0\}.$$

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Now suppose $x \in S$. (This is our usual inductive assumption).

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Then 2 divides $x(x+1)$, so $x(x+1) = 2k$ for some integer k.

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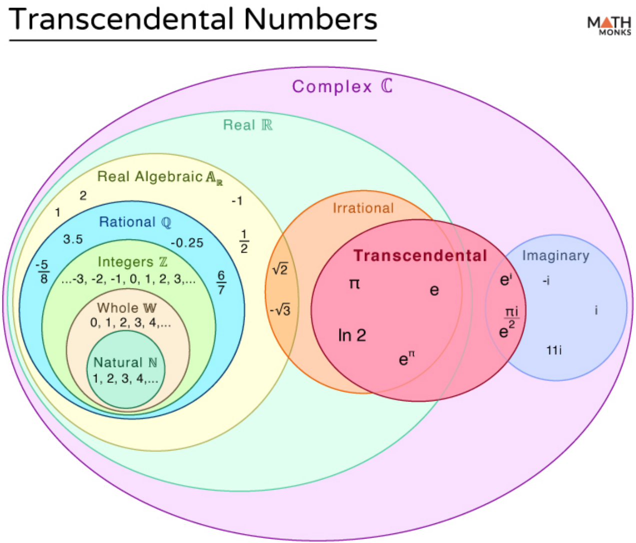
In other words $n \in S$ for every positive integer n.
Hence 2 divides $n(n+1)$ for every positive integers n.

Types of numbers



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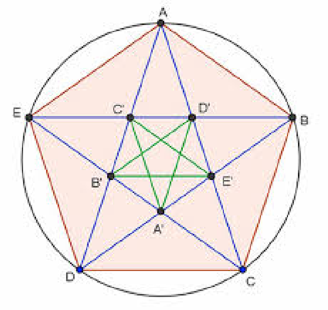
- Discovered by Hippasus, a pythagorean (a student of Pythagoras)



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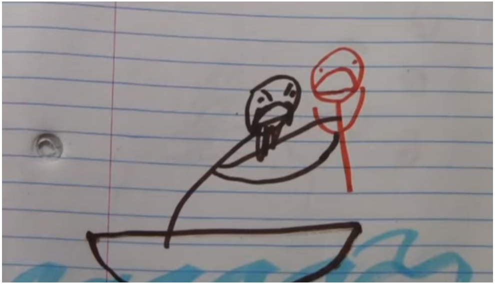

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3. transcendentals are mysterious ... but most real numbers are transcendental!

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Completeness Axiom

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- π is the supremum of

$$\{3, 3.1, 3.14, 3.141, 3.1415, 3.14159, 3.141592, \dots\}.$$

Challenge

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Prove that if A is a nonempty set of integers that is bounded above, then A has a maximum.

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Solution

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Therefore $k \leq 0$ and $a' \leq a$.

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Since b is an upper bound, $b \geq a + k > b - 1 + k$, making $0 > k - 1$.
Therefore $k \leq 0$ and $a' \leq a$.
It follows that a is an upper bound of A, and since $a \in A$ it is a maximum.

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In other words

an infimum is a **greatest lower bound**

Properties of Suprema

The first property of suprema is that they must be arbitrarily close to elements of the set.

Theorem (Approximation Property)

Let $S \subseteq \mathbb{R}$ be nonempty and bounded above, and let $b = \sup(S)$. Then for all $\epsilon > 0$, there exists $a \in S$ with

$$b - \epsilon < a \leq b.$$

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If we remember Taylor series

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$$\sup(S) = 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \dots$$

If we remember Taylor series

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$$

Important example: The number e

Even if ϵ is very small ($\epsilon = 0.000000001$), $\sup(S) - \epsilon$ is not an upper bound of S.
So there exists N with $s_N > \sup(S) - \epsilon$.

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We've just shown that $e^1 = e$ exists.

Properties of Suprema

Also, suprema play nicely with addition.

Theorem (Additive Property)

Let $A, B \subseteq \mathbb{R}$ be nonempty, bounded above sets.

$$C = \{x + y : x \in A,\ y \in B\}.$$

Then C is bounded above and

$$\sup(C) = \sup(A) + \sup(B).$$

Properties of Suprema

Proof.

Let $a = \sup(A)$, $b = \sup(B)$, and $c = \sup(C)$.

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First, note a is an upper bound of A, so $x \leq a$ for all $x \in A$.

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If $z \in C$, then $z = x + y$ for some $x \in A$ and $y \in B$, and therefore $z = x + y \leq a + b$.

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Therefore $a + b$ is an upper bound of C.

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Next, note for all $x \in A$ and $y \leq B$ that $x \leq c - y$.

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Therefore $c - y$ is an upper bound of A and $a \leq c - y$.

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Therefore $c - y$ is an upper bound of A and $a \leq c - y$.
It follows that $y \leq c - a$ for all $y \in B$.

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It follows that $y \leq c - a$ for all $y \in B$.
Thus $c - a$ is an upper bound of B, and it follows $b \leq c - a$.

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It follows that $y \leq c - a$ for all $y \in B$.
Thus $c - a$ is an upper bound of B, and it follows $b \leq c - a$.
Therefore $a + b \leq c$. $\square$

Challenge

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Use the completeness axiom to prove that the $\mathbb{Z}_+$ is not bounded above. (Apostol Theorem 1.17)

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The number b is an upper bound and if $b' < b$, then b' is not.

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The number b is an upper bound and if $b' < b$, then b' is not.
Then $b - 1 < b$ by Axiom 7, so $b - 1$ cannot be an upper bound.

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This means there exists $n \in \mathbb{Z}_+$ with $b - 1 < n$.
It follows from Axiom 7 that $b < n + 1$.

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The number b is an upper bound and if $b' < b$, then b' is not.
Then $b - 1 < b$ by Axiom 7, so $b - 1$ cannot be an upper bound.
This means there exists $n \in \mathbb{Z}_+$ with $b - 1 < n$.
It follows from Axiom 7 that $b < n + 1$.
However, $n + 1 \in \mathbb{Z}$, so this contradicts b being an upper bound.

Archimedian property

Theorem (Apostol Theorem 1.18)

For every $x \in \mathbb{R}$, there exists $n \in \mathbb{Z}_+$ with $n > x$.

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If not, then x is an upper bound of $\mathbb{Z}_+$. □

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Theorem (Archimedian Property of Reals)

For every $x, y \in \mathbb{R}$ with $x > 0$, there exists $n \in \mathbb{Z}_+$ with $y < nx$.

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Proof.

Replace x with y/x in the previous theorem. □